The Berezinskii-Kosterlitz-Thouless Transition

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Abstract

We investigate the Berezinskii-Kosterlitz-Thouless (BKT) topological phase transition in the 2D XY model via Monte Carlo simulations using the Wolff cluster algorithm on varying lattice sizes $(L \times L)$. We leverage parallelism and just-in-time (JIT) compilation to improve computational performance. Thermodynamic observables relevant to the BKT transition, including spin stiffness (helicity modulus), correlation decay exponent η, correlation length ξ, and vortex density, are measured. The critical temperature T_{KT} is estimated from these observables through finite-size scaling analysis employing weighted fits to the form $T_{KT}(L) = T_{KT}(\infty) + a/(\ln L)^2$. The resulting estimates for $T_{KT}(\infty)$ derived from spin stiffness and the $\eta = 1/4$ criterion are found to be in good agreement with the literature value $T_{KT} \approx 0.89213(10)$ (Olsson, 1995) while a notable discrepancy is incurred in the T_{KT} estimate derived from the correlation length ξ. We discuss significant underlying uncertainties inherent to this estimation method which could explain this discrepancy.

1. Introduction

The 2D XY model, a cornerstone in statistical mechanics, describes classical spins confined to a plane on a 2D lattice. Its Hamiltonian is given by

$$H = -J \sum_{\langle i,j \rangle} \cos(\theta_i - \theta_j) \tag{1}$$

(with $J = 1, k_B = 1$ for simplicity in this work), where J is the coupling constant and θ_i is the angle of the spin at site i. This model is pivotal for understanding phase transitions in low-dimensional systems with continuous $O(2)$ symmetry. A crucial aspect of such systems is illuminated by the *Mermin-Wagner theorem* (Mermin & Wagner, 1966), which rigorously precludes spontaneous symmetry breaking and thus conventional long-range order (LRO) at any finite temperature $T > 0$. This theorem initially suggested that 2D systems with continuous symmetries might not exhibit any phase transitions.

However, the 2D XY model famously displays a unique type of phase transition, the *Berezinskii-Kosterlitz-Thouless* (BKT) *transition*, independently predicted by Berezinskii (Berezinskii, 1971; 1972) and Kosterlitz and Thouless (Kosterlitz & Thouless, 1973). This is not a conventional order-disorder transition characterized by a local order parameter and spontaneous symmetry breaking, but rather a topological phase transition of infinite order. It is driven by the binding and unbinding of topological defects known as *vortices* and *antivortices*. Below a *critical temperature* T_{KT}, the system exhibits *quasi-long-range order* (QLRO), where correlations decay algebraically with distance. Above T_{KT}, these vortex-antivortex pairs unbind, proliferate, and screen interactions, leading to a disordered phase with exponentially decaying correlations. The BKT transition has profound implications and finds experimental relevance in diverse physical systems, including thin superfluid films (Nelson & Kosterlitz, 1977), bilayer fluid membranes Park & Lubensky (1996), and 2D superconductors (Zhao et al., 2013).

Contributions. In this course project we satisfy the following three objectives: (1) to investigate the BKT transition in the 2D XY Model via Monte Carlo methods. To this end, we provide a simple but efficient implementation of the Wolff cluster algorithm (Wolff, 1989) and leverage parallelism and just-in-time (JIT) compilation to improve computational performance; (2) to measure thermodynamic observables relevant to the BKT transition, including the spin stiffness (helicity modulus), correlation decay exponent η, and correlation length ξ; (3) to implement finite-size scaling analyses in order to estimate the critical temperature T_{KT} of the BKT transition in the 2D XY Model from these observables. Code available at: `https://github.com/BenBullinger/BKT-Transition-in-the-XY-Model`.

2. Theoretical Background

The BKT transition is characterized by distinct behaviors in spin-spin correlations, the dynamics of topological defects (vortices), and the system's response to an applied twist

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(spin stiffness). We provide a brief overview of the theoretical background based on (Kosterlitz & Thouless, 1973; Kosterlitz, 1974; Berche et al., 2002; Drouin-Touchette, 2022). For the lay reader, we particularly recommend the accessible introduction in Drouin-Touchette (2022).

2.1. Correlation Functions and Universal Exponents

The nature of order in the 2D XY model is captured by the *spin-spin correlation function* $g(r)$, which measures the degree of alignment between two spins separated by a distance r:

$$g(r) = \langle \cos(\theta_i - \theta_j) \rangle. \tag{2}$$

In the low-temperature phase ($T < T_{KT}$), true long-range order is absent, but the system exhibits *quasi-long-range order* (QLRO). This is characterized by an algebraic (power-law) decay of $g(r)$:

$$g(r) \sim r^{-\eta(T)} \tag{3}$$

where $\eta(T)$ is a temperature-dependent critical exponent. The *correlation length* ξ is effectively infinite in this phase. In contrast, in the high-temperature disordered phase ($T > T_{KT}$), correlations decay exponentially:

$$g(r) \sim e^{-r/\xi(T)}. \tag{4}$$

Here, $\xi(T)$ is the finite correlation length, which diminishes as temperature increases further above T_{KT}. A key prediction of BKT theory is that at the critical temperature T_{KT}, the exponent η takes a universal value (Kosterlitz, 1974):

$$\eta(T_{KT}) = 1/4. \tag{5}$$

This provides a powerful method for numerically locating T_{KT}. The correlation length $\xi(T)$ in the disordered phase is predicted to diverge in a non-power-law fashion as T approaches T_{KT} from above (Kosterlitz & Thouless, 1973):

$$\xi(T) \sim \exp(b/\sqrt{T/T_{KT} - 1}) \tag{6}$$

where b is a non-universal constant. This essential singularity is a hallmark of the BKT transition.

2.2. Vortices and Unbinding

Topological defects are crucial to understanding the BKT transition. In the 2D XY model, these are point-like *vortices* and *antivortices*. A vortex is a configuration where the spins wind by a multiple of 2π around a central point (a plaquette center in a lattice model). The *winding number*, or *vorticity*, q is defined as:

$$q = \frac{1}{2\pi} \oint_{\square} \mathrm{d}\theta \tag{7}$$

where the integral is taken around a closed loop $\square$ enclosing the defect. For elementary vortices, $q = \pm 1$ (Drouin-Touchette, 2022).

Kosterlitz and Thouless argued that the transition is driven by an energy-entropy balance for these vortices. The energy of an isolated vortex in a system of size L with lattice spacing a scales as $E_v \approx \pi J \ln(L/a)$. The entropy associated with placing such a vortex in the system scales as $S_v \approx 2k_B \ln(L/a)$, as there are $(L/a)^2$ possible locations. The free energy of an isolated vortex is thus approximately:

$$F_v = E_v - TS_v \approx (\pi J - 2k_B T) \ln(L/a). \tag{8}$$

For $T < T_c = \pi J/(2k_B)$, $F_v > 0$ and diverges with L, meaning isolated vortices are suppressed. For $T > T_c$, $F_v < 0$, and it becomes entropically favorable for free vortices to proliferate, destroying the QLRO (Drouin-Touchette, 2022).

However, this simple argument for isolated vortices is modified by the interaction between vortices. Vortices with opposite winding numbers ($+1$ and -1) attract each other with a logarithmic potential, $U(r) \sim 2\pi J \ln(r/a)$. At low temperatures ($T < T_{KT}$), vortices exist primarily as tightly bound vortex-antivortex pairs, which have a finite energy and do not disrupt QLRO over long distances. As the temperature increases, these pairs begin to unbind. The BKT transition at T_{KT} (which is close to, but not exactly, the T_c from the isolated vortex argument due to renormalization effects) marks the point where it becomes favorable for these pairs to dissociate into free vortices, leading to the disordered phase (Drouin-Touchette, 2022).

2.3. Spin Stiffness and Universal Jump

The *spin stiffness*, also known as the *helicity modulus*, ρ_s, quantifies the system's rigidity or resistance to an imposed long-wavelength twist in the spin configuration. It can be thought of as the free energy cost associated with a phase gradient. In the QLRO phase, ρ_s is finite, indicating the system can support such a gradient. In the disordered phase, ρ_s vanishes in the thermodynamic limit as free vortices screen the twist.

A remarkable prediction of BKT theory, established by Nelson and Kosterlitz (Nelson & Kosterlitz, 1977), concerns the behavior of the spin stiffness at the transition. In the thermodynamic limit ($L \to \infty$), the renormalized spin stiffness $\rho_s(T)$ is finite for $T < T_{KT}$. At the critical temperature T_{KT}, its value when approached from $T < T_{KT}$ is exactly:

$$\lim_{T \to T_{KT}^-} \rho_s(T) = \frac{2k_B T_{KT}}{\pi J}. \tag{9}$$

For any temperature $T > T_{KT}$, the spin stiffness becomes zero in the thermodynamic limit, i.e., $\rho_s(T) = 0$. This discontinuous drop from the value $2k_B T_{KT}/(\pi J)$ to 0 as the temperature crosses T_{KT} is known as the universal jump. For $J = 1, k_B = 1$ (as used in our simulations), the critical value simplifies to $\rho_s(T_{KT}^-) = 2T_{KT}/\pi$. In numerical

simulations on finite-sized systems, this universal relation provides a highly accurate method for estimating T_{KT}: the size-dependent transition temperature $T_{KT}(L)$ is identified as the point where the measured stiffness $\rho_s(T, L)$ for a system of size L intersects the line $y = 2T/\pi$ (Drouin-Touchette, 2022).

2.4. Finite-Size Scaling

Numerical simulations are performed on finite-sized systems, whereas phase transitions are strictly defined in the thermodynamic limit ($L \to \infty$). *Finite-size scaling* (FSS) theory provides the tools to extrapolate results from finite L to this limit. For the BKT transition, the characteristic exponential divergence of the correlation length (6) leads to scaling corrections that are typically logarithmic in L, rather than power-law corrections seen in conventional second-order phase transitions. The size-dependent critical temperature $T_{KT}(L)$, estimated from criteria like the stiffness jump or $\eta(T_{KT}(L)) = 1/4$, is expected to scale towards the thermodynamic limit $T_{KT}(\infty)$ as:

$$T_{KT}(L) = T_{KT}(\infty) + \frac{a}{(\ln L)^2} \tag{10}$$

where a is a non-universal constant. This FSS form is crucial for obtaining precise estimates of $T_{KT}(\infty)$ from simulations on a range of lattice sizes L (Drouin-Touchette, 2022).

3. Numerical Methods and Thermodynamic Observables

We use Monte Carlo (MC) simulations to investigate the BKT transition in the 2D XY Model. We provide implementations of the standard Metropolis algorithm and the Wolff cluster algorithm but restrict the focus of this report to the latter owed to its greater computational efficiency. We further improve computational performance by leveraring parallelization and just-in-time (JIT) compilation of computationally intensive routines using `Numba` (Lam et al., 2015). This section details these algorithms and implementation strategies, alongside the measured thermodynamic observables and statistical error analysis techniques employed.

3.1. Monte Carlo Algorithms

Two principal Monte Carlo update algorithms were implemented: the local *Metropolis-Hastings* algorithm (Metropolis et al., 1953) and the non-local *Wolff* cluster algorithm (Wolff, 1989).

3.1.1. Metropolis Algorithm

The standard single-spin Metropolis algorithm (Metropolis et al., 1953) was implemented as a baseline. For each Metropolis update attempt:

1. A random spin site (i, j) is selected.
2. A new trial angle $\theta_{\text{new}} \in [0, 2\pi)$ is proposed for this spin.
3. The change in energy, $\Delta E = E_{\text{new}} - E_{\text{old}}$, resulting from this proposed move is calculated. Only interactions involving the chosen spin are considered.
4. The move is accepted if $\Delta E \leq 0$. If $\Delta E > 0$, the move is accepted with probability

$$\mathbb{P}_{\text{accept}} = \exp(-\beta \Delta E) \tag{11}$$

where $\beta = 1/(k_B T)$ (with $k_B = 1$ in our simulations).

One Metropolis sweep consists of $N = L \times L$ such update attempts, where L is the linear size of the square lattice. While simple, this local algorithm can suffer from critical slowing down near phase transitions.

3.1.2. Wolff Cluster Algorithm for the XY Model

To more efficiently simulate the system, especially near the BKT transition temperature, the Wolff cluster algorithm (Wolff, 1989) was chosen as the primary update scheme. The steps for one cluster update are:

1. **Random Reflection Axis:** A random angle $\phi_r \in [0, 2\pi)$ is chosen. This defines a reflection axis

$$\mathbf{r} = (\cos \phi_r, \sin \phi_r) \tag{12}$$

2. **Seed Site Selection:** A random spin $\mathbf{S}_k$ at site k (with original angle θ_k) is selected as the seed of the cluster.
3. **Cluster Growth and Reflection:** The chosen seed spin $\mathbf{S}_k$ is immediately reflected with respect to the random axis $\mathbf{r}$ (i.e., its angle θ_k becomes $2\phi_r - \theta_k \pmod{2\pi}$) and is added to a list of spins to be processed. The algorithm then iteratively grows the cluster: A spin $\mathbf{S}_i$ (which has already been reflected and added to the cluster) is taken from the list. For each of its neighbors $\mathbf{S}_j$ not yet in the cluster, a decision is made whether to add $\mathbf{S}_j$. This decision is based on the interaction energy between the already-reflected $\mathbf{S}_i$ (with angle θ_i') and the original state of $\mathbf{S}_j$ (with angle θ_j), compared to the interaction if $\mathbf{S}_j$ were also reflected (to θ_j'). A common way to express the addition probability, ensuring detailed balance, considers the components of the spins projected onto the reflection axis $\mathbf{r}$. Let $s_i = \mathbf{S}_i \cdot \mathbf{r}$ and $s_j = \mathbf{S}_j \cdot \mathbf{r}$. Spin $\mathbf{S}_j$ is added to the cluster with probability:

$$\mathbb{P}_{\text{add}}(i, j) = 1 - \exp[\min(0, 2\beta J s_i s_j)] \tag{13}$$

If $\mathbf{S}_j$ is added, it is immediately reflected ($\theta_j \to 2\phi_r - \theta_j \pmod{2\pi}$) and added to the list of spins to be processed. This process continues until the list is empty, at which point the cluster is fully formed and all its members have been reflected.

One Wolff sweep consists of $N = L \times L$ such cluster construction attempts, each starting from a new random seed spin.

3.2. Implementation Details

Several strategies were employed for robust and performant simulations:

- **Just-In-Time (JIT) Compilation:** Computationally intensive functions, including energy calculations, Monte Carlo update steps (Metropolis and Wolff), and observable measurements, were implemented with just-in-time compilation using `Numba` (Lam et al., 2015).
- **Precomputation of Neighbor Indices:** We implement the lattices with periodic boundary conditions (PBC). We precompute the four nearest neighbors for each site and store them at initialization.
- **Simulation Protocol:** For each lattice size L and temperature T, the system was initialized with random spin orientations. It was then equilibrated for a specified number of equilibration/thermalization sweeps. Following equilibration, a specified number of Monte Carlo sweeps were performed for measurements. One MC sweep constitutes N Metropolis attempts or $N = L \times L$ Wolff cluster constructions.
- **Parallelization:** Simulations for different (L, T) pairs were distributed across multiple CPU cores.

3.3. Observables and Error Analysis

We measured a range of thermodynamic observables. We quantified the measurement errors using the standard error for direct averages and using Jackknife resampling for derived quantities and fitted parameters.

3.3.1. Measured Observables

- **Energy (e):** Average energy per spin, calculated from

$$H = -J \sum_{\langle i,j \rangle} \cos(\theta_i - \theta_j) \tag{14}$$

Reported as $\langle E/N \rangle$.

- **Specific Heat (c_v):** Calculated from energy fluctuations:

$$c_v = (\langle E^2 \rangle - \langle E \rangle^2)/(NT^2) \tag{15}$$

- **Magnetization ($\langle |m| \rangle$):** Average magnitude of the total magnetization vector per spin.

$$\mathbf{M} = \left(\sum_k \cos\theta_k, \sum_k \sin\theta_k \right) \tag{16}$$

$m = |\mathbf{M}|/N$. Reported as $\langle m \rangle$.

- **Susceptibility (χ):** From magnetization fluctuations:

$$\chi = N(\langle m^2 \rangle - \langle m \rangle^2)/T \tag{17}$$

- **Binder Cumulant (U_L):**

$$U_L = 1 - \langle m^4 \rangle / (3\langle m^2 \rangle^2) \tag{18}$$

- **Spin Stiffness (ρ_s):** The helicity modulus, ρ_s, quantifies the system's resistance to an imposed twist. For an isotropic system on a square lattice, it is typically calculated by averaging the stiffness components along the principal axes. Thus,

$$\rho_s = (\rho_{s,x} + \rho_{s,y})/2 \tag{19}$$

The x-component, $\rho_{s,x}$, specifically measures the stiffness against a twist in the x-direction. It is calculated from terms involving sums over all bonds oriented purely in the x-direction. These terms are:

$$S_{cx} = \sum_{\text{x-direction bonds}} J \cos(\theta_i - \theta_j) \tag{20}$$

$$S_{sx} = \sum_{\text{x-direction bonds}} J \sin(\theta_i - \theta_j) \tag{21}$$

where the sums are over all pairs of nearest-neighbor spins (i, j) connected horizontally. These sums are computed for each instantaneous configuration. Then, $\rho_{s,x}$ is derived from averages over many configurations:

$$\rho_{s,x} = \frac{1}{N} \left(\langle S_{cx} \rangle - \beta \langle (S_{sx})^2 \rangle \right) \tag{22}$$

A similar expression holds for $\rho_{s,y}$ (using sums over y-direction bonds).

- **Vortex Density (ω_v):** Average absolute vorticity per plaquette. For a plaquette,

$$q_p = \frac{1}{2\pi} \sum_{\text{bonds}} \text{wrap}(\theta_j - \theta_i) \tag{23}$$

where $\text{wrap}(x) = ((x + \pi) \pmod{2\pi}) - \pi$. Reported as $\omega_v = \langle |q_p| \rangle$.

- **Spin-Spin Correlation Function ($g(r)$):**

$$\langle \cos(\theta_i - \theta_j) \rangle \tag{24}$$

versus separation r, measured up to $r = L/2$.

- **Correlation Exponent (η) and Length (ξ):** Extracted by fitting $g(r)$ (averaged over MC steps). We define $T_{\text{fit_boundary}} \approx 1.1$. Then
  - For $T < T_{\text{fit_boundary}}$: use
  $$g(r) = Ar^{-\eta(T)} \tag{25}$$
  to determine $\eta(T)$.
  - For $T > T_{\text{fit_boundary}}$: use
  $$g(r) = A\exp(-r/\xi(T)) \tag{26}$$
  to determine $\xi(T)$.

3.3.2. Error Analysis Methods

Statistical errors are inherent in Monte Carlo simulations. In this study, two primary methods were employed:

- **Standard Error of the Mean (SEM):** For quantities that are direct averages of N_{meas} measurements from sufficiently decorrelated configurations the *Standard Error of the Mean* (SEM)
  $$\sigma_{\text{SEM}} = s/\sqrt{N_{\text{meas}}} \tag{27}$$
  was used, where s is the sample standard deviation of the N_{meas} observations (computed with $N_{\text{meas}} - 1$ degrees of freedom). Specifically, SEM is used for the energy per spin $\langle e \rangle$ and the average magnetization magnitude $\langle |m| \rangle$)
- **Jackknife Resampling:** For more complex quantities derived from averages of non-linear functions, those involving fluctuations, or parameters obtained from fitting, *Jackknife resampling* was employed. The procedure is as follows:
  1. Given N_{meas} original measurements (e.g., N_{meas} values of total energy E_k, or N_{meas} instantaneous correlation functions $g_k(r)$).
  2. Create N_{meas} Jackknife samples. The k-th Jackknife sample is formed by omitting the k-th original measurement from the dataset.
  3. The observable of interest, $\mathcal{O}$, is recalculated for each of these N_{meas} Jackknife samples, yielding $\mathcal{O}_{(k)}$. For η and ξ, this involves re-fitting the averaged correlation function (computed from $N_{\text{meas}} - 1$ raw $g(r)$ functions) for each Jackknife sample.
  4. The Jackknife estimate of the variance of $\mathcal{O}$ is then given by:
  $$\sigma^2_{\text{JK}}(\mathcal{O}) = \frac{N_{\text{meas}} - 1}{N_{\text{meas}}} \sum_{k=1}^{N_{\text{meas}}} (\mathcal{O}_{(k)} - \bar{\mathcal{O}}_{(\cdot)})^2 \tag{28}$$
  where $\bar{\mathcal{O}}_{(\cdot)} = \frac{1}{N_{\text{meas}}} \sum_{k=1}^{N_{\text{meas}}} \mathcal{O}_{(k)}$ is the mean of the Jackknife estimates. The Jackknife error is $\sigma_{\text{JK}}(\mathcal{O})$.

  Specifically, Jackknife resampling is used for the specific heat (c_v), susceptibility (χ), Binder cumulant (U_L), spin stiffness (ρ_s), vortex density (ω_v), and the fitted correlation parameters (η, ξ)

4. Results and Discussion

We carried out simulations using the Wolff cluster algorithm for lattice sizes $L \in \{20, 30, 40, 50, 60, 70\}$ with $J = 1$ and $k_B = 1$. For $L = 20$ we provide a visualization of spin configurations at different temperatures around T_{KT} in Fig. 1.

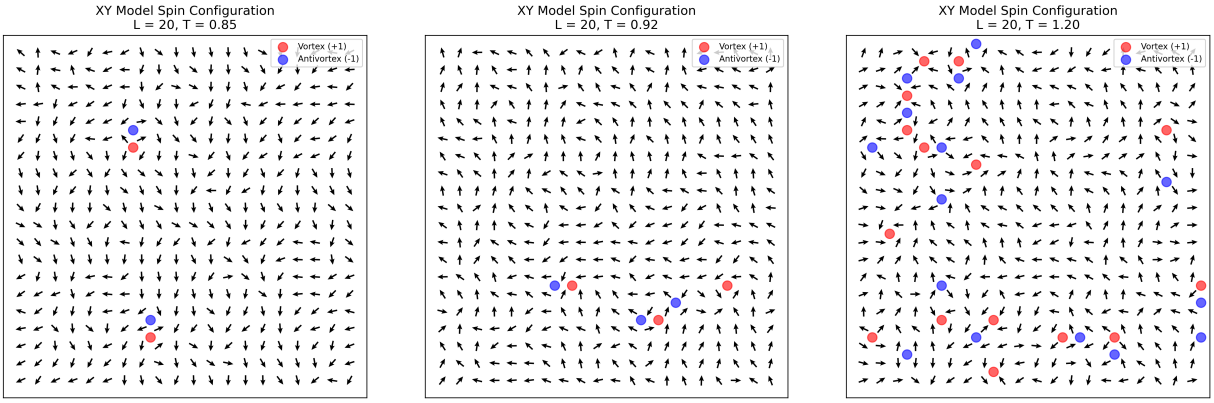
4.1. Qualitative Observations of Spin Configurations

Visual inspection of spin configurations, as exemplified in Fig. 1, reveals key features of the BKT transition. At low temperatures (e.g., $T = 0.85$, Fig. 1(a)), the system exhibits quasi-long-range order with predominantly bound vortex-antivortex pairs. Near T_{KT} (e.g., $T = 0.92$, Fig. 1(b)), these pairs begin to unbind. At high temperatures (e.g., $T = 1.2$, Fig. 1(c)), a proliferation of free vortices destroys long-range correlations, leading to a disordered phase. These observations align with established BKT theory.

4.2. Analysis of Thermodynamic Observables

We present the measurements of the thermodynamic observables for different lattice sizes in Fig. 2. Their apparent behavior is as follows:

- **Energy (e)** (Fig. 2(a)): Increases monotonically with increasing temperature. No sharp discontinuity is observed.
- **Specific Heat (c_v)** (Fig. 2(b)): Exhibits a broad, smooth peak around $T \approx 1.0 - 1.1$. This peak does not diverge with system size L, a hallmark of the infinite-order BKT transition.
- **Magnetization ($\langle |m| \rangle$)** (Fig. 2(c)): Suppressed for $T > 0$ and vanishes as $L \to \infty$, consistent with the Mermin-Wagner theorem. The decay with L is slower in the QLRO phase.
- **Susceptibility (χ)** (Fig. 2(d)): Shows a peak that grows with system size L. For $T \leq T_{KT}$, $\chi(T)$ is expected to diverge in the thermodynamic limit.
- **Binder Cumulant (U_L)** (Fig. 2(e)): Transitions from ≈ 1 at low T towards $1/3$ at high T. Intersections of $U_L(T)$ curves for different L appear less sharp than for conventional critical points.



(a) Low T ($T = 0.85 < T_{KT}$) (b) Near T_{KT} ($T = 0.92 \approx T_{KT}$) (c) High T ($T = 1.2 > T_{KT}$)

Figure 1. Spin configurations and vortex locations ($L = 20$). The arrows represent spins while red and blue circles show vortices and antivortices, respectively. Note the bound vortex-antivortex pairs at low temperature (a), their progressive unbinding near T_{KT} (b), and free vortices at high temperature (c).

- **Vortex Density** (ω_v) (Fig. 2(f)): Remains low at low temperatures (bound pairs) and increases sharply around T_{KT} due to the unbinding and proliferation of free vortices.

Error bars in these plots, derived from SEM or Jackknife resampling as implemented (see Section 3.3.2), quantify the statistical uncertainty.

4.3. Finite-Size Scaling Analysis

Measurements for the three primary indicators for the BKT transition (spin stiffness ρ_s, correlation exponent η, and correlation length ξ) are presented in Fig. 3. Their corresponding finite-size scaling (FSS) analyses, used to extrapolate the transition temperature T_{KT} to the thermodynamic limit, are shown in Fig. 4.

In order to discuss the FSS analyses we first describe how the finite-size transition temperatures $T_{KT}(L)$ are determined from the indicators ρ_s, η, and ξ.

Spin Stiffness (ρ_s): As shown in Fig. 3(a), the spin stiffness $\rho_s(T)$ is a primary indicator. For each system size L, the estimate $T_{KT}^{\rho_s}(L)$ is found where $\rho_s(T, L)$ intersects the Nelson-Kosterlitz line $2T/\pi$. We perform linear interpolation and propagate the Jackknife error of $\rho_s(T, L)$ to determine the error on $T_{KT}^{\rho_s}(L)$.

Correlation Exponent (η): Fig. 3(b) displays $\eta(T)$. For each L, $T_{KT}^{\eta}(L)$ is estimated as the temperature where the fitted $\eta(T, L)$ equals its universal BKT value of $1/4$. We interpolate this crossing point and propagate the Jackknife error of $\eta(T, L)$ to obtain the error on $T_{KT}^{\eta}(L)$.

Correlation Length (ξ): Fig. 3(c) shows $\xi(T)$. For temperatures T above an initial estimate of $T_{KT}(L)$ (obtained e.g. from the η criterion), we perform a weighted non-linear fit of the BKT divergence form in Eq. (6) to the measured $\xi(T, L)$, using the Jackknife errors of $\xi(T, L)$ as weights. The error on the extracted $T_{KT}^{\xi}(L)$ is derived from the covariance matrix of this fit.

Finite-Size Scaling Extrapolation: The series of size-dependent critical temperatures $T_{KT}^{\rho_s}(L)$, $T_{KT}^{\eta}(L)$, and $T_{KT}^{\xi}(L)$ are extrapolated to the thermodynamic limit ($L \to \infty$) according to the scaling form in Eq. (10).

We perform a weighted linear fit of the scaling form in Eq. (10), using the propagated errors of $T_{KT}(L)$ (i.e. $\sigma_{T_{KT}(L)}$) as weights. The y-intercept of this fit yields the estimate for $T_{KT}(\infty)$ and its statistical error is computed as the covariance matrix of the fit parameters. This FSS procedure is applied independently to the three sets of $T_{KT}(L)$ estimates $T_{KT}^{\rho_s}(L)$, $T_{KT}^{\eta}(L)$, and $T_{KT}^{\xi}(L)$. The resulting plots are shown in Fig. 4.

We obtain the following results for $T_{KT}(\infty)$:

- Spin Stiffness ρ_s (Fig. 4(a)):

$$T_{KT}^{\rho_s}(\infty) \approx 0.8934(24)$$

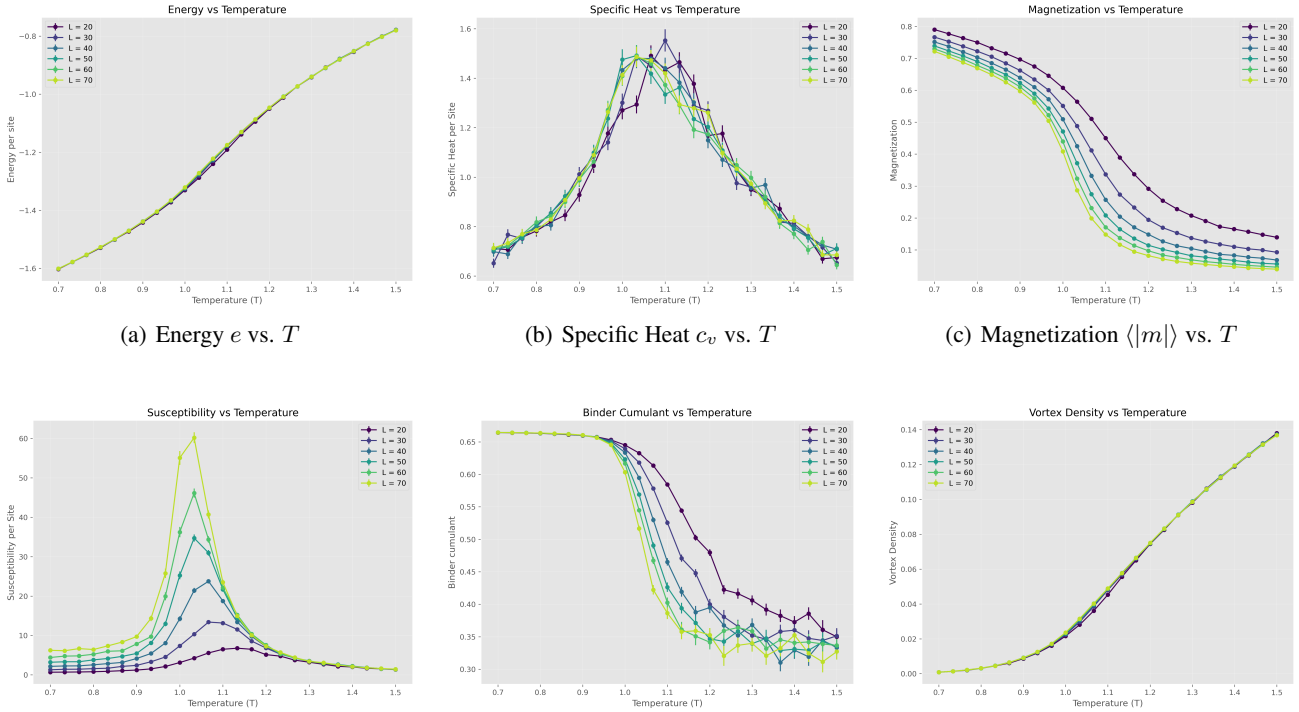
- Correlation exponent η (Fig. 4(b)):

$$T_{KT}^{\eta}(\infty) \approx 0.8946(22)$$

- ξ Correlation length ξ (Fig. 4(c)):

$$T_{KT}^{\xi}(\infty) \approx 0.8098(73)$$

While only the estimate $T_{KT}^{\rho_s}(\infty)$ is consistent with the literature value of $T_{KT} \approx 0.89213(10)$ (Olsson, 1995) in



(a) Energy e vs. T (b) Specific Heat c_v vs. T (c) Magnetization $\langle|m|\rangle$ vs. T

(d) Susceptibility χ vs. T (e) Binder Cumulant U_L vs. T (f) Vortex Density ω_v vs. T

Figure 2. General thermodynamic observables and vortex density. (a) Energy, (b) Specific Heat, (c) Magnetization, (d) Susceptibility, (e) Binder Cumulant, and (f) Vortex Density. Error bars are shown where calculated (refer to Section 3.3.2 for error estimation methods).

terms of the determined confidence intervals, we notice that the estimate $T_{KT}^{\eta}(\infty)$ is also close to the literature value. This is expected as the spin stiffness method – directly probing the Nelson-Kosterlitz prediction – and the $\eta = 1/4$ criterion – a fundamental characteristic of the BKT phase – are generally considered robust estimators of the BKT transition temperature.

The estimate from the correlation length divergence, $T_{KT}^{\xi}(\infty) \approx 0.8098(73)$, is notably lower and does not align with the literature value or the other two methods. This discrepancy is not entirely unexpected. The determination of T_{KT} from fitting the essential singularity form of $\xi(T)$ (Eq. (6)) is known to be highly sensitive to several factors: the precise temperature range used for the fit (which must be carefully chosen above the true T_{KT}), the initial parameter guesses for the non-linear fit (including the non-universal constant b), and the statistical quality of the $\xi(T, L)$ data itself, which is obtained from a prior fit of $g(r)$. These factors can lead to larger systematic uncertainties or biases in the $T_{KT}^{\xi}(L)$ estimates compared to those from ρ_s or η.

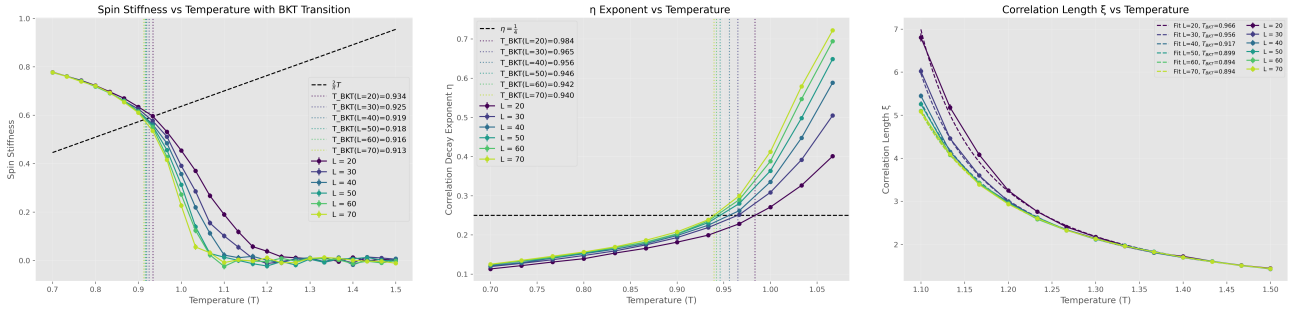
Considering these observations, the results obtained from spin stiffness and the $\eta = 1/4$ criterion provide the most reliable estimates of the BKT transition temperature in this study, consistent with established findings.

5. Conclusion

We successfully investigated the Berezinskii-Kosterlitz-Thouless (BKT) transition in the 2D XY model through comprehensive Monte Carlo simulations. Our study focused on the Wolff cluster algorithm to efficiently simulate systems of varying lattice sizes. A suite of thermodynamic observables was measured, including energy, specific heat, magnetization, susceptibility, Binder cumulant, spin stiffness, vortex density, and the spin-spin correlation function, from which the exponent η and length ξ were derived.

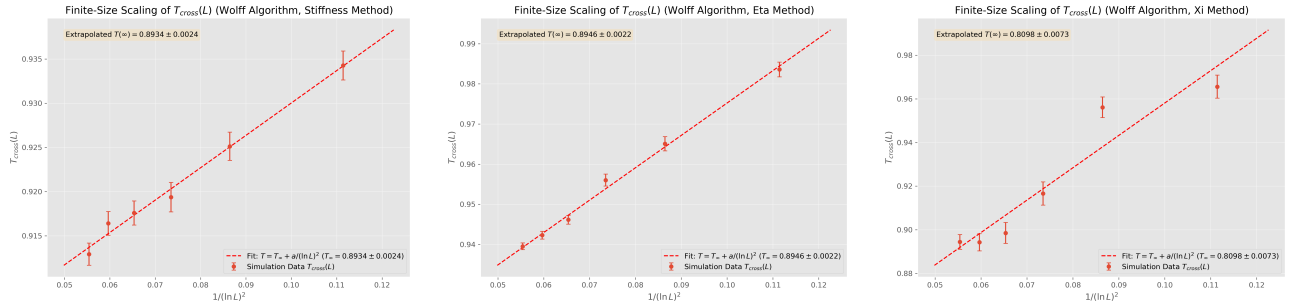
Size-dependent critical temperatures, $T_{KT}(L)$, were determined using three established criteria: the Nelson-Kosterlitz universal jump in spin stiffness (ρ_s), the correlation exponent achieving its universal value ($\eta = 1/4$), and the characteristic BKT divergence of the correlation length (ξ). These $T_{KT}(L)$ values, with their propagated errors, were extrapolated to the thermodynamic limit ($L \to \infty$) via finite-size scaling.

Our analysis yielded estimates for $T_{KT}(\infty)$ from these three methods. The values obtained from spin stiffness ($T_{KT}^{\rho_s}(\infty) \approx 0.8934(24)$) and the correlation exponent



(a) Spin Stiffness ρ_s vs. T (b) Exponent η vs. T (c) Correlation Length ξ vs. T

Figure 3. Key BKT transition indicators. (a) Spin Stiffness ρ_s (dashed line is $2T/\pi$), (b) Correlation exponent η (dashed line is $\eta = 1/4$), (c) Correlation length ξ. Error bars are shown where calculated using methods described in Section 3.3.2.



(a) FSS for ρ_s-derived $T_{KT}(L)$ (b) FSS for η-derived $T_{KT}(L)$ (c) FSS for ξ-derived $T_{KT}(L)$

Figure 4. Finite-size scaling (FSS) analyses for $T_{KT}(L)$ derived from the indicators in Fig. 3: (a) stiffness, (b) $\eta = 1/4$, and (c) ξ fitting. Plots are against $1/(\ln L)^2$, with weighted linear fits extrapolating to $T_{KT}(\infty)$.

($T^{\eta}_{KT}(\infty) \approx 0.8946(22)$) were found to be in good agreement with the high-precision literature value of $T_{KT} \approx 0.89213(10)$ (Olsson, 1995). The estimate derived from the correlation length divergence ($T^{\xi}_{KT}(\infty) \approx 0.8098(73)$) showed a notable deviation, underscoring the known sensitivities of this particular fitting procedure.

In conclusion, this project successfully replicated key features of the BKT transition and demonstrated the effective application of advanced Monte Carlo techniques, statistical error handling, and appropriate finite-size scaling analysis. Our results obtained from the spin stiffness and η criterion broadly agree with the established results in the literature. However, the estimate derived from the correlation length ξ highlights the importance of careful methodological choices in numerical studies of phase transitions.

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